Chao (Charles) Cheng

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EDUCATION

Ph.D. in Finance (expected), Zhongnan University of Economics and Law 2023–2027
Advisor: Jinlin Zhang

M.Econ. in Finance, Hainan University 2020–2023

B.Eng. in Information Security, Hefei University of Technology 2011–2015

WORK EXPERIENCE

Overseas brand planning, JAC Motors International Co. 2015–2019

RESEARCH INTERESTS

Economics of artificial intelligence; Computational economics (agent-based models, reinforcement learning); Behavioral finance and futures markets; Household finance and policy evaluation

PUBLICATIONS

1. "When Artificial Intelligence Generates and Experts Validate: Specialist Allocation and the Aggregate Substitutability of Artificial Intelligence in Research" (with Jinlin Zhang). *Systems, forthcoming*.
2. "When Pilots Overlap: Contamination Bias in Difference-in-Differences Evaluations of China's City Pilot Policies for Sustainable Development" (with Jinlin Zhang). *Sustainability, forthcoming*.
3. "Cost-Based Price Floors, Soft Budgets and Capacity Adjustment in China's Solar Photovoltaic Industry" (with Jinlin Zhang). *Sustainability, 2026, 18(21): 11136*.
4. "The Existence of Anchoring Effect of Chinese Commodity Futures Investors and Its Influence on Market Efficiency" (with Xiong Kuang and Xia Chen). *Journal of Systems Science and Mathematical Sciences, 2022, 42(8): 2126–2141 (in Chinese)*.

PAPERS UNDER REVIEW

1. "Generative AI Adoption and Back-Office Employment: Early Evidence from Chinese Listed Firms" (with Jinyun Pan). *Under review, China Economic Review*.
2. "Do Below-Cost Sales Bans Facilitate Algorithmic Collusion? Price Floors, Learning and Cost-Based Regulation". *Under review, Computational Economics*.
3. "Taking Instructions at Their Word: Price Floors, Prompt Protocols and LLM Pricing Agents". *Under review, Journal of Behavioral and Experimental Economics*.
4. "Pooled Margin Accounts and Cross-Product Transmission in Futures Markets: Account Evidence and a Calibrated Agent-Based Model". *Under review, Journal of Futures Markets*.
5. "Who Gets Liquidated? Margin Requirements and the Reallocation of Liquidation Risk in Futures Markets: An Agent-Based Systems Analysis" (with Jinlin Zhang). *Under review, Systems*.
6. "The Efficiency and Equity of Trade-in Subsidies: Evidence from the China Household Finance Survey" (with Jinlin Zhang). *Under review, Public Finance Research*.

WORKING PAPERS

1. "Fleeing or flocking? Generative AI and the demand for college majors in China". *Working paper*.

2. “Bias or Patience? The Disposition Effect, Trading Function, and Pricing Efficiency in Futures Markets” (with Jinlin Zhang). *Working paper*.
3. “Do Managers Underreact? Subjective Uncertainty and the Revision of Range Earnings Forecasts”. *Working paper*.
4. “Housing Space, Purchasing Capacity, and the Upgrading of Household Durable Goods: Evidence from the China Household Finance Survey” (with Jinlin Zhang). *Working paper (in Chinese)*.
5. “Monitoring and Early Warning of Systemic Financial Risk: The Evolution of Methods and Their Applicability in China”. *Working paper (in Chinese)*.

BOOKS

1. *Agent-Based Modeling in Economic Research: From Introduction to Practice*. *Book manuscript (in Chinese)*.
2. *Capital Markets and Common Prosperity: Wealth Creation, Distribution and Governance* (with Jinlin Zhang). *Book manuscript (in Chinese)*.

SOFTWARE

- **llmarket**: An open-source agent-based simulation toolkit for economics, with LLM agents, reinforcement learners and price regulation. Developer and maintainer, Python, MIT License, v0.1.2, October 2026. github.com/CharlesChengGithub/llmarket

RESEARCH PROJECTS

Major Program of the National Social Science Fund of China, “Financial Theory and Effective Mechanisms for Advancing Common Prosperity” (22&ZD115); participant. Related output: the book *Capital Markets and Common Prosperity* — Since 2023

China Futures Association Joint Research Program (Class II project), “Investor Behavioral Factors in China’s Futures Market and Their Effects on Price Discovery”; team member, ranked second. Responsible for data collection, analysis and the research report — 2021–2022

“Risk Prevention and Regulation of Offshore Finance in the Hainan Free Trade Port”; participant. Reviewed international experience with regulatory sandboxes and drafted a framework adapted to Hainan — 2021

TEACHING INTERESTS

Finance; Investments; FinTech; Behavioral Finance; Econometrics; Python Programming; Financial Data Analysis and Visualization; introductory computing and AI literacy courses

SKILLS

Programming	Python (pandas, NumPy; agent-based modeling, reinforcement learning, LLM agent experiments), Java, Stata, NetLogo; Jupyter Notebook, Git and GitHub, LaTeX
Data	Cleaned and analyzed nearly five million futures brokerage records (trades, closed positions and daily holdings) in Python; China Household Finance Survey (CHFS), CSMAR and annual-report text, gaokao admission data
Methods	Agent-based modeling, Q-learning, LLM agents, DSGE, Monte Carlo simulation, difference-in-differences and panel econometrics
Languages	Chinese (native), English (CET-6, 538)

Last updated: November 2026